

Problem 2: Quadrature

Heun's method for the ODE $y'(t) = f(y)$ is

$$\begin{aligned}k_1 &= f(y_n) \\k_2 &= f(\tilde{y}_{n+1}) = f(y_n + hk_1) \\y_{n+1} &= y_n + \frac{h}{2} [k_1 + k_2]\end{aligned}$$

where h is the step size. The ODE can also be written as an integral equation

$$y(t) = y(0) + \int_0^t f(y(s))ds$$

(a) Explain how a single step of Heun's rule can be thought of as a left-endpoint quadrature followed by an approximate Trapezoid rule quadrature.

(b) Use the quadrature error formula for the simple Trapezoid rule, together with Taylor series, to derive a bound on the error in one step of Heun's method. You may assume that $y_n = y(t_n)$, i.e. there is no error in y_n .

Solution: **(a)** The first stage of Heun's rule (which is a second order Runge Kutta method) is a left-endpoint quadrature (equivalent to explicit Euler)

$$y(t_{n+1}) \approx \tilde{y}_{n+1} = y_n + hf(y_n).$$

The simple trapezoid rule approximation of the integral over $[t_n, t_n + h]$ is

$$y(t_n + h) = y_n + \int_{t_n}^{t_n+h} f(y(s))ds \approx y_n + \frac{h}{2} [f(y_n) + f(y(t_n + h))].$$

Since the exact value $y(t_n + h)$ is unknown, Heun's rule uses the approximation $y(t_n + h) \approx \tilde{y}_{n+1}$ to produce

$$y(t_n + h) \approx y_n + \frac{h}{2} [f(y_n) + f(\tilde{y}_{n+1})] = y_n + \frac{h}{2} [f(y_n) + f(y_n + hf(y_n))].$$

(b) For the error bound we need to account for the trapezoid rule error and the error associated with replacing $y(t_n + h)$ by $\tilde{y}_{n+1}$. To develop a formula for the error in $\tilde{y}_{n+1}$ we just need a Taylor expansion

$$y(t_n + h) = y_n + hf(y_n) + \frac{h^2}{2} f'(y(\xi))y'(\xi)$$

for some $\xi \in [t_n, t_n + h]$.

From this we find

$$|e_1| = |y(t_n + h) - \tilde{y}_{n+1}| \leq \frac{h^2}{2} \|f'(y(t))y'(t)\|_\infty.$$

For the simple trapezoid rule the error is

$$y(t_n + h) = y_n + \frac{h}{2} (f(y_n) + f(y(t_n + h))) - \frac{h^3}{12} \left. \frac{d^2 f(y(t))}{dt^2} \right|_{t=\zeta}$$

where ζ is an unknown value $\zeta \in [t_n, t_n + h]$. We use Taylor series to account for the error associated with replacing $y(t_n + h)$ by $\tilde{y}_{n+1}$:

$$\begin{aligned}
y(t_n + h) &= y_n + \frac{h}{2} (f(y_n) + f(\tilde{y}_{n+1} + e_1)) - \frac{h^3}{12} \frac{d^2 f(y(t))}{dt^2} \Big|_{t=\zeta} \\
&= y_n + \frac{h}{2} (f(y_n) + f(\tilde{y}_{n+1}) + e_1 f'(\psi)) - \frac{h^3}{12} \frac{d^2 f(y(t))}{dt^2} \Big|_{t=\zeta} \\
&= y_{n+1} + \frac{he_1}{2} f'(\psi) - \frac{h^3}{12} \frac{d^2 f(y(t))}{dt^2} \Big|_{t=\zeta} \\
\Rightarrow y(t_n + h) - y_{n+1} &= \frac{he_1}{2} f'(\psi) - \frac{h^3}{12} \frac{d^2 f(y(t))}{dt^2} \Big|_{t=\zeta}.
\end{aligned}$$

where ψ is some unknown value of y between y_n and $y(t_n + h)$.

Using the bound above on $|e_1|$ and the triangle inequality we obtain

$$|y(t_n + h) - y_{n+1}| \leq h^3 \left[\frac{\|f'(y(t))y'(t)\|_\infty}{2} + \frac{1}{12} \left\| \frac{d^2[f(y(t))]}{dt^2} \right\|_\infty \right].$$

Since Heun's method is locally third order, it is globally second order.

Problem 3: Numerical Linear Algebra

The basic QR iteration for finding the eigenvalues of a real matrix A is

$$A_{m-1} = Q_{m-1}R_{m-1}, \quad A_m = R_{m-1}Q_{m-1}, \quad A_0 = A$$

where $A_{m-1} = Q_{m-1}R_{m-1}$ is the QR factorization of A_{m-1} .

- (a) Prove that the eigenvalues of A_m are the same as the eigenvalues of A .
- (b) Explain how to construct an upper Hessenberg matrix H that has the same eigenvalues as A .
- (c) Prove that if A is upper Hessenberg, then A_m is also upper Hessenberg. (You may assume that A is invertible, and that Q_m is upper Hessenberg whenever A_m is upper Hessenberg.)

Solution: (a) Since

$$A_m = Q_{m-1}^T A_{m-1} Q_{m-1}$$

A_m is similar to A_{m-1} so they must have the same eigenvalues. An inductive step proves that it is true for all $m \geq 1$.

(b) Let the first column of A be

$$\begin{pmatrix} a_{1,1} \\ \mathbf{a}_{2:n,1} \end{pmatrix}.$$

Let

$$\mathbf{u}_1 = \frac{\mathbf{a}_{2:n,1} - \|\mathbf{a}_{2:n,1}\|_2 \mathbf{e}_1}{\|\mathbf{a}_{2:n,1} - \|\mathbf{a}_{2:n,1}\|_2 \mathbf{e}_1\|_2}$$

and define

$$H_1 = \left[\begin{array}{c|c} 1 & \mathbf{0}^T \\ \hline \mathbf{0} & I - 2\mathbf{u}_1\mathbf{u}_1^T \end{array} \right].$$

Note that

$$H_1 A H_1^T = \left[\begin{array}{c|c} * & *^T \\ \hline \|\mathbf{a}_{2:n,1}\|_2 & A^{(2)} \\ 0 & \end{array} \right].$$

If we recursively apply this idea to $A^{(2)}$ using a matrix H_2 of the form

$$H_2 = \left[\begin{array}{cc|c} 1 & 0 & \mathbf{0}^T \\ 0 & 1 & \mathbf{0}^T \\ \hline \mathbf{0} & \mathbf{0} & I - 2\mathbf{u}_2\mathbf{u}_2^T \end{array} \right]$$

etc., then we can see that the final result

$$H = H_{n-2} \cdots H_1 A H_1^T \cdots H_{n-2}^T$$

will be upper Hessenberg.

Since the H_i are all orthogonal matrices this is a similarity transform, so H has the same eigenvalues as A .

(c) This is true even if A is not invertible, but the proof for invertible matrices is easier. Start with

$$A = Q_0 R_0.$$

Next note that

$$A_1 = R_0 Q_0 = R_0 A R_0^{-1}.$$

The lower bandwidths of the matrices in the rightmost expression are 0, 1, and 0. Since multiplication adds the bandwidths, the product on the right has lower bandwidth ≤ 1 , meaning that it is upper Hessenberg. An inductive step shows that A_m will be upper Hessenberg for all $m \geq 1$.

Problem 4: Interpolation/Approximation

(a) Find the quadratic that interpolates the following temperature data: $T(-1) = 4$, $T(0) = 10$, $T(1) = 20$.

(b) Suppose that density is related to temperature via $\rho(T) = \rho_0 - \alpha T$, and that the total mass is known to be m

$$\int_{-1}^1 \rho(T(x)) dx = m.$$

Find a cubic polynomial that interpolates the data from (a) and satisfies the above integral constraint, or explain why none exists. For part (b) let $\rho_0 = 1/2$, $m = 1$, and $\alpha = 1$.

(c) Consider the problem of both interpolating the data and satisfying the integral constraint for an arbitrary set of $n + 1$ distinct interpolation nodes $x_0, \dots, x_n$ using a polynomial of degree $\leq n + 1$. When a solution exists, it must have the form

$$p(x) = q(x) + \sum_{j=0}^n T(x_j) \ell_j(x).$$

- (i) When a solution does exist, give an explicit formula for $q(x)$.
- (ii) Give an explicit criterion for when a solution does not exist.

Solution: (a) The solution is

$$p(x) = 10 + 8x + 2x^2.$$

(b) One way to approach this is to let

$$p(x) = \sum_{j=0}^3 a_j x^j$$

and then solve the following system

$$\begin{bmatrix} 1 & -1 & 1 & -1 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 2 & 0 & \frac{2}{3} & 0 \end{bmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} 4 \\ 10 \\ 20 \\ -\frac{m-2\rho_0}{\alpha} \end{pmatrix}.$$

If you row reduce you will end up with a final equation of the form

$$0 = -\frac{64}{3}$$

so no solution exists.

(c)

- (i) q must have the form

$$q(x) = \lambda \ell(x)$$

where $\ell(x) = \prod_{j=0}^n (x - x_j)$ is the node polynomial. This is the only possible solution for a polynomial q of degree at most $n + 1$ that satisfies $q(x_j) = 0$ for $j = 0, \dots, n$.

The above is not a complete solution; we still need λ . The integral constraint is

$$\lambda \int_{-1}^1 \ell(x) dx + \sum_{j=0}^n T(x_j) \int_{-1}^1 \ell_j(x) dx = -\frac{m - 2\rho_0}{\alpha}.$$

The formula for λ is

$$\lambda = -\frac{1}{\int_{-1}^1 \ell(x) dx} \left[\frac{m - 2\rho_0}{\alpha} + \sum_{j=0}^n T(x_j) \int_{-1}^1 \ell_j(x) dx \right].$$

(ii) It is tempting to say that a solution does not exist when

$$\int_{-1}^1 \ell(x) dx = 0$$

which happened in part (b). But this is only half the answer. If, by chance, we have

$$\sum_{j=0}^n T(x_j) \int_{-1}^1 \ell_j(x) dx = -\frac{m - 2\rho_0}{\alpha}$$

then $\lambda = 0$ is an answer regardless of the value of $\int_{-1}^1 \ell(x) dx$. In this case there are an infinite number of solutions. So the full criterion for when a solution does not exist is

$$\int_{-1}^1 \ell(x) dx = 0 \textbf{ and } \sum_{j=0}^n T(x_j) \int_{-1}^1 \ell_j(x) dx \neq -\frac{m - 2\rho_0}{\alpha}.$$

Problem 5: Numerical ODE

Consider the boundary value problem

$$-\frac{d}{dx} \left(a(x) \frac{du}{dx} \right) = f(x), \quad u(0) = u(1) = 0$$

where $a(x) > \delta \geq 0$ is a bounded differentiable function in $[0, 1]$. We note that the above ODE can be written as

$$-\frac{da}{dx} \frac{du}{dx} - a(x) \frac{d^2u}{dx^2} = f(x), \quad u(0) = u(1) = 0.$$

We assume that, although $a(x)$ is available, an expression for its derivative, $\frac{da}{dx}$, is not available.

(a) Using finite differences and an equally spaced grid in $[0, 1]$, $x_l = hl$, $l = 0, \dots, n$ and $h = 1/n$, we discretize the ODE to obtain a linear system of equations, yielding an $O(h^2)$ approximation of the ODE. After the application of the boundary conditions, the resulting coefficient matrix of the linear system is an $(n-1) \times (n-1)$ tridiagonal matrix.

Provide a derivation and write down the resulting linear system (by giving the expressions of the elements).

(b) Utilizing all the information provided, find a disc in $\mathbb{C}$, the smaller the better, that is guaranteed to contain all the eigenvalues of the linear system constructed in part (a).

Solution:

(a) We must choose an $O(h^2)$ finite difference approximation for the derivatives. I choose to use centered differences to approximate the derivatives since it is known that

$$\left. \frac{du}{dx} \right|_{x_j} = \frac{u(x_{j+1}) - u(x_{j-1}))}{2h} + O(h^2)$$

and

$$\left. \frac{d^2u}{dx^2} \right|_{x_j} = \frac{u(x_{j+1}) - 2u(x_j) + u(x_{j-1}))}{h^2} + O(h^2).$$

Let u_j denote the approximate value $u(x_j)$. Then truncating the $O(h^2)$ terms of the derivatives and plugging them into the differential equation, we find the following row equation at x_j for $j = 1, \dots, n-1$

$$-\left(\frac{a(x_{j+1}) - a(x_{j-1}))}{4h^2} \right) (u_{j+1} - u_{j-1}) - \frac{a(x_j)}{h^2} (u_{j+1} - 2u_j + u_{j-1}) = f(x_j) \quad (1)$$

Since we know that $u(0) = u(1) = 0$, we get different equations for $j = 1$ and $j = n-1$.

For $j = 1$, the equation is given by

$$-\left(\frac{a(x_2) - a(x_0)}{4h^2} \right) (u_2) - \frac{a(x_1)}{h^2} (u_2 - 2u_1) = f(x_1). \quad (2)$$

For $j = n-1$, the equation is given by

$$-\left(\frac{a(x_n) - a(x_{n-2}))}{4h^2} \right) (-u_{n-2}) - \frac{a(x_{n-1})}{h^2} (-2u_{n-1} + u_{n-2}) = f(x_{n-1}). \quad (3)$$

The resulting tridiagonal system has RHS $f(x_j)$ for $j = 1, 2, \dots, n-1$;

— diagonal entries:

$$2 \frac{a(x_j)}{h^2}, \quad j = 1, \dots, n-1;$$

— lower diagonal entries:

$$\left(\frac{a(x_{j+1}) - a(x_{j-1}))}{4h^2} \right) - \frac{a(x_j)}{h^2} \quad j = 2, \dots, n-1;$$

— upper diagonal entries:

$$- \left(\frac{a(x_{j+1}) - a(x_{j-1}))}{4h^2} \right) - \frac{a(x_j)}{h^2} \quad j = 1, \dots, n-2.$$

(b) The Gershgorin theorem states that for any complex $n \times n$ matrix $\mathbf{A}$, the eigenvalues lie within the collection of disc with radius $R_i = \sum_{j \neq i} |a_{ij}|$ centered at a_{ii} .

Applying this to the matrix, we find that the Gershgorin disc for the linear system are

$$\left| \lambda - \frac{2a(x_1)}{h^2} \right| \leq \left| \frac{a(x_2) - a(0)}{4h^2} + \frac{a(x_1)}{h^2} \right|,$$

$$\left| \lambda - \frac{2a(x_j)}{h^2} \right| \leq 2 \left| \frac{a(x_{j+1}) - a(x_{j-1}))}{4h^2} + \frac{a(x_j)}{h^2} \right|, \quad \text{for } j = 2, \dots, n-2,$$

and

$$\left| \lambda - \frac{2a(x_{n-1})}{h^2} \right| \leq \left| \frac{a(1) - a(x_{n-2})}{4h^2} + \frac{a(x_{n-1})}{h^2} \right|.$$

Now we must find a disc which contains all of these. To do this, we utilize the fact that $a(x) \geq \delta > 0$. Let $M = \max_{x \in [0,1]} |a(x)|$. Then

$$\left| \lambda - \frac{2a(x_j)}{h^2} \right| \leq 2 \left| \frac{a(x_{j+1}) - a(x_{j-1}))}{4h^2} + \frac{a(x_j)}{h^2} \right| \leq \frac{2}{h^2} \left(\frac{M - \delta}{4} + M \right).$$

Now that we have an upper bound for the radius of all the disc, we note that $\delta < a(x_j) \leq M \forall j$. Thus to make sure we capture all eigenvalues, we center the disc at $\frac{\delta + M}{h^2}$ in the middle of $\frac{2\delta}{h^2}$ and $\frac{2M}{h^2}$, then add $\frac{M - \delta}{h^2}$ to the radius we found above.

Finally the smallest disc that we find is

$$\left| \lambda - \frac{M + \delta}{h^2} \right| \leq \frac{M - \delta}{h^2} + \frac{2}{h^2} \left(\frac{M - \delta}{4} + M \right) = \frac{1}{h^2} (3.5M - 1.5\delta)$$

Problem 6: Numerical PDE

Consider the equation

$$\begin{aligned} u_t + au_x &= 0, & a \in \mathbb{R}, t > 0, \\ u(x, 0) &= f(x). \end{aligned} \quad (4)$$

The solution will be approximated using the Finite Difference Lax-Wendroff method

$$v_j^{n+1} = v_j^n - \frac{a\Delta t}{2\Delta x} (v_{j+1}^n - v_{j-1}^n) + a^2 \frac{(\Delta t)^2}{2(\Delta x)^2} (v_{j+1}^n - 2v_j^n + v_{j-1}^n)$$

where $v_j^n = u(x_j, t_n)$ is a grid function, and Δx and Δt denote the spacing between grid points in the x and t directions.

NOTE: That there are two centered difference formulas used in the spatial direction.

(a)

$$\frac{v_j^{n+1} - v_j^n}{\Delta t} = u_t + \frac{\Delta t}{2} u_{tt} + \frac{(\Delta t)^2}{6} u_{ttt} + O((\Delta t)^3)$$

Rewrite this expression replacing the temporal derivatives u_{tt} and u_{ttt} in terms of spatial derivatives using equation (4).

(b) Determine the spatial and temporal orders of accuracy of the Lax-Wendroff method.

(c) Use von Neumann analysis to determine under what conditions the method is stable. (Hint: it is useful to look at the square of the amplification factor.)

Solution:

(a)

$$\frac{v_j^{n+1} - v_j^n}{\Delta t} = u_t + \frac{a^2 \Delta t}{2} u_{xx} + \frac{a^3 (\Delta t)^2}{6} u_{xxx} + O((\Delta t)^3)$$

(b)

$$\begin{aligned} \frac{v_{j+1}^n - v_{j-1}^n}{2\Delta x} &= u_x + \frac{(\Delta x)^2}{6} u_{xxx} + O((\Delta x)^4) \\ \frac{v_{j+1}^n - 2v_j^n + v_{j-1}^n}{(\Delta x)^2} &= u_{xx} + \frac{(\Delta x)^2}{12} u_{xxxx} + O((\Delta x)^4) \end{aligned}$$

Plugging these two Taylor expansions and the solution to part (a) into the finite difference scheme, we get

$$u_t + au_x = -\frac{a^3 (\Delta t)^2}{6} u_{xxx} - \frac{a (\Delta x)^2}{6} u_{xxx} + O((\Delta t)^3) + O(\Delta x)^4$$

So the method is $O((\Delta x)^2 + (\Delta t)^2)$.

(c) We plug $v_j^n = \xi^n e^{ikx_j}$ into the finite difference method.

$$\begin{aligned} v_j^{n+1} &= v_j^n - \frac{a\Delta t}{2\Delta x} (v_{j+1}^n - v_{j-1}^n) + a^2 \frac{(\Delta t)^2}{2(\Delta x)^2} (v_{j+1}^n - 2v_j^n + v_{j-1}^n) \\ \xi^{n+1} e^{ikx_j} &= \xi^n e^{ikx_j} - \frac{a\Delta t}{2\Delta x} (\xi^n e^{ik(x_j+\Delta x)} - \xi^n e^{ik(x_j-\Delta x)}) + a^2 \frac{(\Delta t)^2}{2(\Delta x)^2} (\xi^n e^{ik(x_j+\Delta x)} - 2\xi^n e^{ikx_j} + \xi^n e^{ik(x_j-\Delta x)}) \\ \xi &= 1 - \frac{a\Delta t}{2\Delta x} (e^{ik\Delta x} - e^{-ik\Delta x}) + a^2 \frac{(\Delta t)^2}{2(\Delta x)^2} (e^{ik\Delta x} - 2 + e^{-ik\Delta x}) \\ &= 1 - \frac{a\Delta t}{\Delta x} i \sin(k\Delta x) + a^2 \frac{(\Delta t)^2}{2(\Delta x)^2} (\cos(k\Delta x) - 1) \end{aligned}$$

Let $\kappa = \frac{a\Delta t}{\Delta x}$, then ξ can be written in a slightly cleaner form.

$$\xi = 1 - \frac{a\Delta t}{\Delta x} i \sin(k\Delta x) + \kappa^2 (\cos(k\Delta x) - 1)$$

$$\begin{aligned} |\xi|^2 &= 1 - 2\kappa^2(1 - \cos(k\Delta x)) + \kappa^2(1 - \cos^2(k\Delta x)) + \kappa^4(1 - \cos(k\Delta x))^2 \\ &= 1 - \kappa^2(1 - \kappa^2)(1 - \cos(k\Delta x))^2 \\ &= 1 - 4\kappa^2(1 - \kappa^2) \sin^4\left(\frac{1}{2}k\Delta x\right) \end{aligned}$$

The method is stable when $|\xi|^2 \leq 1$ which matches the CFL condition requiring $|\kappa| \leq 1$.